

Customer Churn EDA Report

(1)Objective

The goal of this analysis is to identify customer behavior patterns associated with churn and discover potential retention opportunities.

(2)Dataset Overview

- Number of customers: 6000
- Number of features: 15
- Target variable: Churn
- Churn percentage:15.5%

The dataset contains customer behavioral and engagement information. The churn rate is approximately 15%, indicating a moderate class imbalance.

(3)Data Quality Check

The dataset was checked for missing values and duplicate records, and no data quality issues were identified.

(4)Customer Activity Analysis

Feature: days_since_last_purchase

Finding:

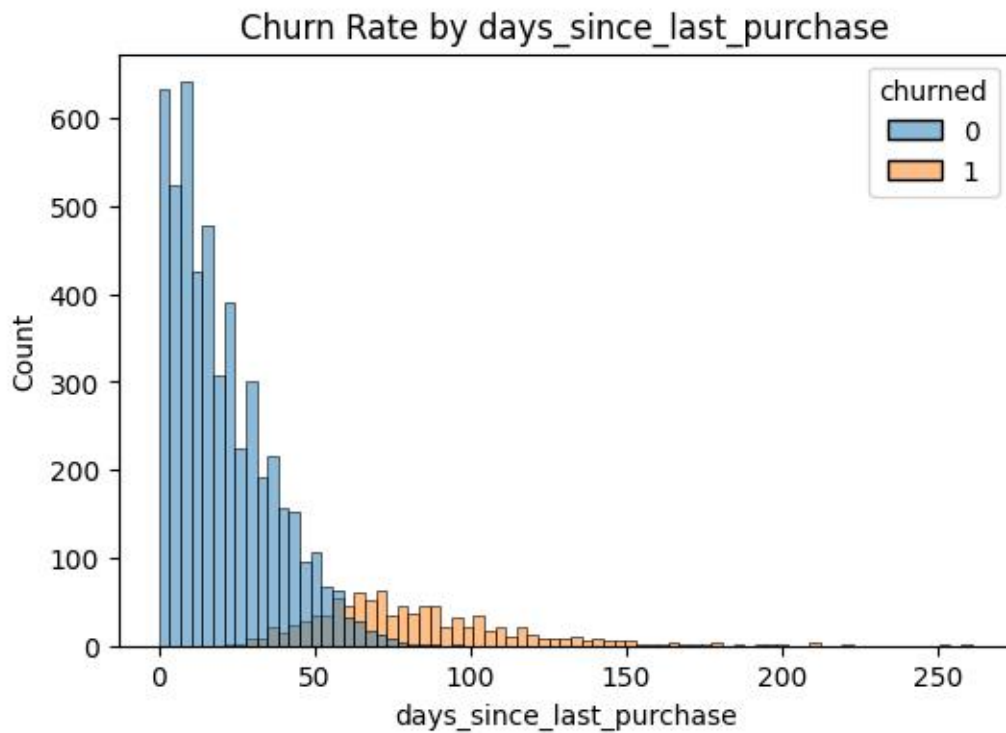
Customers with longer periods since their last purchase show higher churn risk.

Business Interpretation:

Reduced purchase activity may represent an early warning signal for customer disengagement.

Recommendation:

Implement re-engagement campaigns for inactive customers.



(5)Engagement Analysis

Feature: engagement_score

Finding:

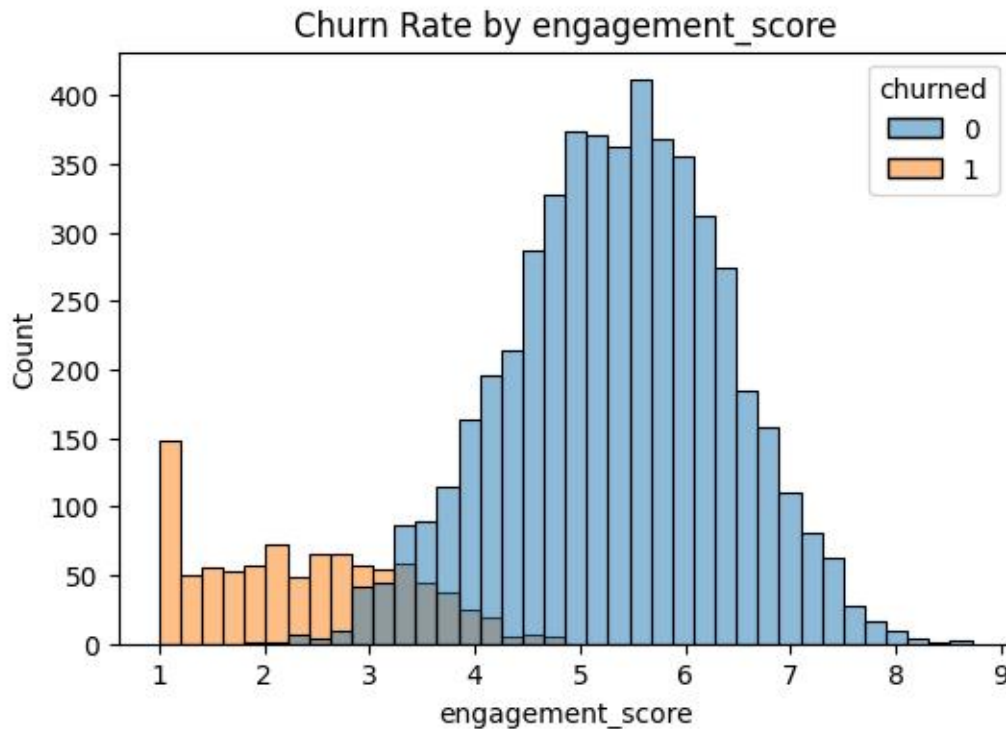
Customers with lower engagement scores are more likely to churn

Business Interpretation:

Maintaining customer interaction is important for retention.

Recommendation:

Create engagement campaigns targeting low-activity customers



(6)Loyalty Analysis

Feature: loyalty_member

Finding:

Loyalty members show lower churn rates compared with non-members.

Business Interpretation:

The loyalty program appears associated with stronger customer retention.

Recommendation:

Encourage more customers to join the loyalty program.

(7)Churn Customer Profile:

A typical churn-risk customer is characterized by:

- Long time since last purchase
- Low engagement level.
- Higher return activity
- Lower satisfaction
- No loyalty membership

(9) Conclusion:

- Customers with longer periods since their last purchase show higher churn risk.
- Customers with lower engagement scores are more likely to churn
- Loyalty members show lower churn rates compared with non-members.